SF1811 Optimization, 2026/2027
Homework assignment 1
Due Tuesday November 10 2026 to qualify for bonus point

General instructions for the homework assignments are given in the course memo. The required ethical approach is also given in the course memo. Additional clarifications are given below.

A homework assignment is carried out in a group consisting of one or two students, where the students select the groups. For each homework assignment, a group should hand in a written report in Canvas. There may be different groups for the different homework assignments.

- Discussion between groups is encouraged, but each group must individually solve the assignment.
- Generative AI tools may be used as support, but the submitted solution must reflect the students' own mathematical reasoning and understanding.

  AI tools may be used, for example, to discuss concepts, obtain feedback on an approach, check language, debug code, or generate plots. AI should primarily support learning and critical review. It must not replace the students' responsibility for formulating, verifying, and presenting the final solution.

  Any use of generative AI must be declared in the submission. The declaration must state:

  - which tools were used;
  - for what purposes they were used;
  - which parts of the submission were affected; and
  - how the generated material was checked, modified, or verified.

  If no generative AI tool was used, this should also be stated.

  For work submitted in pairs, the declaration is joint, and both students are responsible for the complete submission. Each student must be able to explain and justify all mathematical arguments, computations, code, and conclusions without AI assistance.

  Declared use of generative AI does not in itself affect the assessment. The examiner or teaching assistants may ask individual oral follow-up questions to verify understanding. A submission may not be accepted if the students cannot explain or justify its contents.

  Students are expected to declare their use of generative AI honestly and to the best of their knowledge. Deliberately undeclared or misrepresented use may be treated as suspected academic misconduct and handled according to KTH regulations.

- Instructions for the report:

  - The report should have a leading title page where the group members' names, personal number and e-mail addresses are clearly stated. (One or two members.)
  - The report should be written using a suitable word processor. Figures need not be computer generated, but the report should be compiled into one single pdf file.

  - The report should give clear answers to each exercise. The contents should be such that another student who has taken the course, but is not familiar with this particular assignment, should be able to read the report and understand.
  - Matlab code should not be part of the report, and should not be referred to in the report.

- Each group should upload the following documents via the Canvas page of the course no later than by the deadline of the assignment:

  - The report as a single pdf file.
  - Additional Matlab files if applicable, i.e., if you have written your own Matlab files.

  Please upload your documents as individual pdf and m files, and not as zip files.

The intent of this homework assignment is to use the simplex method for solving a linear program in standard form,

$$(LP) \qquad \begin{array}{ll} \underset{x \in \mathbb{R}^n}{\text{minimize}} & c^T x \\ \text{subject to} & Ax = b, \\ & x \geq 0. \end{array}$$

There are Matlab functions `simplex.m` and `polyplot.m` available in Canvas to assist you.

Exercise 1.1 Consider the simplex method, as stated in the formula sheet.

- Expand the terse description in the formula sheet to a brief description of the method, motivating and explaining the different steps of the method in your own words.

Exercise 1.2 Your task is to state and illustrate the problem given by A, b and c.

- Copy the Matlab file `hw1data.m` from Canvas and run
  "`[A,b,c,B] = hw1data(YYMMDD)`",
  where `YYMMDD` should be replaced by one group member's date of birth. Then A is a matrix of dimension 4×6, $B = (3\ 4\ 5\ 6)$ is a basis which gives a basic feasible solution as $A_B = I$ and $b \geq 0$. In addition, as $c_B = 0$, x_3, x_4, x_5 and x_6 are slack variables. The problem may therefore be illustrated in x_1 and x_2 only, eliminating the slack variables.
- State this two-dimensional problem and illustrate it in a figure. The Matlab file `polyplot.m`, which is available in Canvas, may be helpful.

Exercise 1.3 Your task is to solve the linear program given by A, b and c using the simplex method, with the given B as initial basis. You may either (i) write your own simplex solver in Matlab, (ii) do manual calculations in Matlab, or (iii) make use of the Matlab file `simplex.m`, which can be found in Canvas. If you use `simplex.m`, you will have to modify it in order to obtain the required information for each iteration. The notation in `simplex.m` is chosen to match that of the formula sheet, in order to be helpful for your learning. Even if you choose not to make use of `simplex.m` for obtaining your results, make use of it to verify that your x, y and r of each step coincide with those of `simplex.m`.

- In the step of the simplex method, where it is stated "*Otherwise, choose $k \in N$ with $r_k < 0$.*", choose $k \in N$ such that $r_k = \min_{j \in N} r_j$. (This is the choice originally proposed by George Dantzig when developing the method.)

- Describe the first iteration in detail, matching the description of Exercise 1.1.
- For the remaining iterations, it suffices to report B, x, y and r.
- Motivate optimality of your final solution.
- In addition, you should illustrate the x_1 and x_2 values of each iteration in the figure created in Exercise 1.2.

Exercise 1.4 When solving the problem for the given initial B, more than one component of r_N is negative in the first iteration.

- In the first iteration, choose a different k from the one chosen in Exercise 1.3, and otherwise repeat that exercise.
- For each iteration, report r_N and verify that the selected entering index k satisfies $r_k < 0$. If more than one component of r_N is negative, motivate your choice of k. You need not describe all details of the first iteration again.

Good luck!

Below are given two additional exercises. They are not part of the assignment and no solutions should be handed in. They concern useful issues to be covered later in the course. The instructor would be happy to discuss them after class or during office hours with anyone interested.

- It may appear limited that we only solve a problem of low dimension where an initial feasible basis is known. However, the work we have done is directly applicable for problems of larger dimensions where no initial feasible point is known. The Matlab file `simplex12.m`, which is available in Canvas, uses `simplex.m` to perform Phase I and, when a feasible basis has been obtained, Phase II. First, the so-called Phase I problem

$$\begin{array}{ll} \underset{x\in\mathbb{R}^n,u\in\mathbb{R}^m}{\text{minimize}} & \sum_{i=1}^m u_i \\ \text{subject to} & Ax + Du = b, \\ & x \geq 0,\ u \geq 0, \end{array}$$

  is solved, where $D = \text{diag}(d)$, with

$$d_i = \begin{cases} 1, & b_i \geq 0, \\ -1, & b_i < 0. \end{cases}$$

  Then, there is an x which is feasible to (LP) if and only if the optimal value of the Phase I problem is zero. Hence, by solving the Phase I problem, we can determine whether (LP) is feasible and, in the usual case, obtain a feasible basis from which Phase II can be started. See *Optimization, §5.3.*

  Suggestions:

  - Solve the linear program of Exercise 1.2 by "`[x,z,y,r,B,iter1,iter2] = simplex12(A,b,c)`".
  - Add a constraint to your linear program that makes the feasible region empty. Check that `simplex12` correctly verifies infeasibility.
  - Solve a slightly larger problem for which no initial basic feasible solution is known, for example, one give by `m=50;n=200;A=randn(m,n);b=A*ones(n,1);c=rand(n,1);`.

- Consider the perturbed problem

$$(LP_\delta^i) \qquad \begin{array}{ll} \underset{x\in\mathbb{R}^n}{\text{minimize}} & c^T x \\ \text{subject to} & Ax = b + \delta e_i, \\ & x \geq 0, \end{array}$$

  where e_i is the ith unit vector and δ is a scalar. Let x^*, y^* and r^* be the values of x, y and r associated with an optimal solution of (LP). In addition, let z^* denote the optimal value of (LP) and let z_δ^i denote the optimal value of (LP_δ^i).

  Claim: *It holds that $z_\delta^i = z^* + y_i^*\delta$ for δ close to zero (positive and negative); and $z_\delta^i \geq z^* + y_i^*\delta$ for all δ, (where we interpret the optimal value as infinity for an infeasible problem).*

  This means that we can actually know the optimal value of (LP_δ^i) for values of δ close to zero without solving (LP_δ^i).

  Suggestions:

– Verify that this is the case for your example problem by solving (LP_δ^i) for some i and some values of δ close to zero. In your instructor's case, with "`[A,b,c,B] = hw1data(610729)`", the choice of $i = 2$ gives

$$z_\delta^2 = -11.6572 - 1.8614\delta,$$

which is valid for $-1.0890 \leq \delta \leq 0.3817$. (The bounds on δ for which equality is valid require additional calculations and cannot be obtained directly from the solution of (LP).)

– Can you explain?